PHYSICAL CHEMISTRY

GUIDED REVISION PLAN-SCORE ADVANCED

Topic : Thermodynamics & Thermochemistry (Question Bank # 04)

MM : 72 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. Each of the vessel 1 and 2 contain 1.2 moles of gaseous helium. The ratio of the vessel volumes is $V_2/V_1 = 2.0$, and the ratio of the absolute temperature of helium in them is $T_1/T_2 = 2.0$. Assuming the gas to be ideal, find the difference of gas entropies in these vessels, $S_2 - S_1$. (ln 2 = 0.7).

(A) 0.84 cal/ K (B) 4.2 cal/ K

(C) –0.84 cal/ K (D) – 4.2 cal/ K

2. The entropy change accompanying the transfer of 12000 J of heat from a body A at 327ºC to a body B at 127ºC is -

(A) –10.0 J/K (B) +10.0 J/K (C) –57.8 J/K (D) +57.8 J/K

3. $2NO_2(g) \rightarrow N_2O_4(g)$

$\Delta U^\circ_f [N_2O_4(g)] = 2$kcal/ mole

and $\Delta U^\circ_{reaction} = -16$ kcal /mol then calculate $\Delta H^\circ_{formation}$ of NO_2 (g) at 727ºC

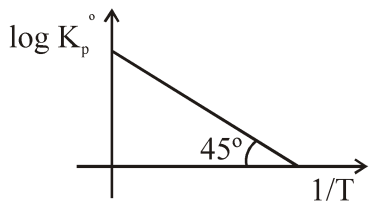
(A) 9 kcal /mol (B) 4.5 kcal /mol

(C) 8 kcal /mol (D) 10 kcal /mol

4. For a process which follows the equation $PV^3 = C$, the work done when one mole of ideal gas was taken from 1 atm to $2\sqrt{2}$ atm starting from initial temperature of 300 K is

(A) 300 R (B) 150 R (C) 600 R (D) 900 R

5. For a reaction graph of log K_p° and 1/T as follows -



then standard enthalpy change for reaction will be -

(A) 2 cal/mol (B) –2 cal/mol (C) 4.606 cal/mol (D) –4.606 cal/mol

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6. Which one of the following statement is false.

(A) For a fix amount of a substance entropy of a substance in the liquid phase will be lower than the entropy of the same substance in gaseous phase

(B) Spontaneous process always occur very rapidly

(C) for a fix amount of a solid ΔS sublimation of a solid always positive

(D) If $\Delta G < 0$, then it is not necessary that process will be always spontaneous

7. Heat liberated by a given amount of an ideal gas undergoing reversible isothermal process is 1200 cal at 300K. What is the Gibbs free energy change of the gas in this process ?

(A) zero

(B) +1200 cal

(C) –1200 cal

(D) 4 cal

8. ΔH_f^0 of water is – 285.8 kJ mol^{-1}. If enthalpy of neutralisation of monoacid strong base is –57.3 kJ mol^{-1}, ΔH_f^0 of OH^- (aq.) ion will be

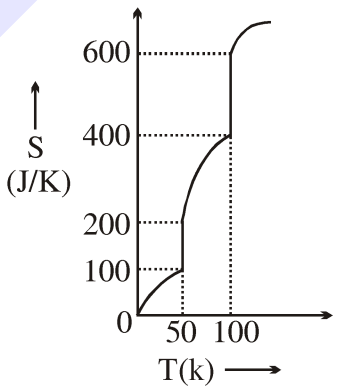
(A) – 228.5 kJ mol^{-1}

(B) 228.5 kJ mol^{-1}

(C) 114.25 kJ mol^{-1}

(D) –114.25 kJ mol^{-1}

9. For 1 mole substance variation of absolute entropy with temperature is given by following graph. What is **molar enthalpy of vapourisation (ΔH_{vap}).**



(A) 2 kJ

(B) 5 kJ

(C) 20 kJ

(D) 10 kJ

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(ii) One or more options correct Type

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

10. Suppose that the volume of a certain ideal gas is to be doubled by one of the following processes

(A) isothermal expansion

(B) adiabatic expansion

(C) free expansion in insulated condition

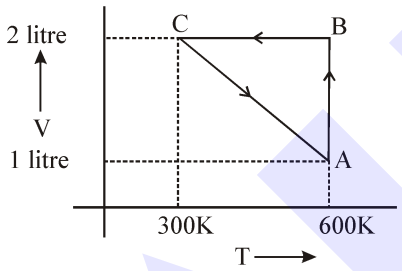
(D) expansion at constant pressure.

If E_1,E_2,E_3 and E_4 respectively are the changes in average kinetic energy of the molecules for the above four processes, then-

(A) $E_2 = E_3$ (B) $E_1 = E_3$

(C) $E_1 > E_4$ (D) $E_4 > E_3$

11. 1 mole $N_2(g)$ undergo following cyclic process -



which option (s) is/are correct - [Use : R = 2 Cal / K - mole ; ln 2 = 0.7]

(A) $W_{AB} = -840$ cal (B) $\Delta H_{AB} = 300$ cal

(C) $\Delta U_{CA} = 1500$ cal (D) $q_{BC} = -600$ cal.

12. For $MX_2(s)$ lattice enthalpy is 800kJ mol^{-1} and enthalpy of solution is 50kJ mol^{-1}. If hydration energies of M^{2+} and X^- ions are in the ratio 3 : 1 , then correct options are :

(A) Hydration energy of M^{2+} is –300 kJ/mol

(B) Hydration energy of M^{2+} is –450 kJ/mol

(C) Hydration energy of X^- is –300 kJ/mol

(D) Hydration energy of X^- is –150 kJ/mol

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(iv) Table Type

- This section contains **1** question of matching type.
- This section contains **ONE** table (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : –1 In all other cases

Column-I **(Reactions)**	**Column-II** **(Characteristics)**	**Column-III** **(Characteristics)**
(I) $C(s) + O_2(g) \rightarrow CO_2(g)$	(i) ΔH = +ve	(P) $\Delta H > \Delta E$
(II) $SO_3(g) \rightarrow SO_2(g) + \frac{1}{2}O_2(g)$	(ii) Δn_g = +ve	(Q) $\lvert \Delta H \rvert > \lvert \Delta E \rvert$
(III) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$	(iii) Δn_g = –ve	(R) $\Delta H < \Delta E$
(IV) $C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$	(iv) ΔH = –ve	(S) $\lvert \Delta H \rvert < \lvert \Delta E \rvert$

13. Which of the following is only correct match ?

(A) I , i , P (B) II , ii , Q

(C) III , iii , S (D) IV , iv , R

14. Which of the following is only incorrect match ?

(A) II , i , P (B) III , iv , R

(C) IV , iii , S (D) II , ii , Q

15. Which of the following is only correct match ?

(A) I , ii , Q (B) II , iii , Q

(C) III , iv , S (D) IV , ii , P

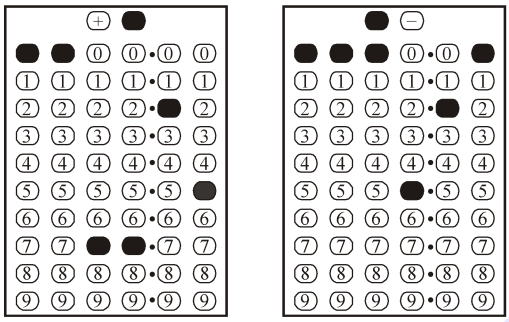
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SECTION-II: (Integer Second Decimal)

- This section contains **4** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

  For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +4 If ONLY the correct numerical value is entered as answer.

  Zero Marks : 0 In all other cases.

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1. At 500 kilobar pressure, density of diamond and graphite are 3 g/cc and 2 g/cc respectively, at certain temperature 'T'. Find the value $|\Delta H - \Delta U|$ (kJ/mole) for the conversion of 1 mole of graphite to 1 mole of diamond at temperature 'T' :

2. A solution of 8.4 gm of a newly discovered compound (molar mass = 80000) in 25 ml of solution shows a temperature rise of 0.04ºC for complete oxygenation. Each mole of compound binds 10 moles of oxygen. If heat capacity of solution is 4.2 J/K- ml then calculate amount of heat released per mole of oxygen bound in kJ

3. The increase in Gibbs free energy (in kJ) of 13 g of ethanol (density = 0.78 g cm^3), when the pressure is increased isothermally from 1 bar to 3001 bar, is

4. The entropy change for the isothermal reversible expansion of an ideal gas at 300K from a particular state A to a state B is 20 J/K. Calculate the work (in kJ) obtained in the process.

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SECTION–IV : Matrix-Match Type

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. Match the column

	Column–I		**Column–II**
(A)	Vaporisation of water at 100°C & 1 atm	(P)	$\Delta S > 0$
(B)	Isothermal reversible expansion of an ideal gas	(Q)	$\Delta H = 0$
(C)	Adiabatic free expansion of ideal gas	(R)	$\Delta G < 0$
(D)	Isobaric heating of an ideal gas	(S)	$W = 0$
		(T)	$q > 0$

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ANSWER KEY

SECTION–I : (i) Only One option correct Type : 3(–1)

1.	Ans. (C)	2.	Ans. (B)	3.	Ans. (C)
4.	Ans. (B)	5.	Ans. (C)	6.	Ans. (B)
7.	Ans. (B)	8.	Ans. (A)	9.	Ans. (C)

(ii) One or more options correct Type : 4(–1)

10.	Ans. (B,D)	11.	Ans. (A,C)	12.	Ans. (B, D)

(iv) Table Type :

13.	Ans. (B)	14.	Ans. (C)	15.	Ans. (D)

SECTION-II: (Integer Second Decimal) : 4(0)

1.	Ans. 100.00	2.	Ans. (4.00)	3.	Ans. (5.00)
4.	Ans. (6.00)				

SECTION–IV : Matrix-Match Type : 8(0)

1. Ans. (A)–P, T ; (B)–P,Q,R,T ; (C)–P,Q,R,S ; (D)-P,R,T

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